

Notes on Canonical Transformations

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Feb. 16, 2026

These notes are intended as supporting material for my video *Canonical Transformations: the bridge to quantum mechanics* [1].

1 Simplifying Kepler's problem

Kepler's problem is the task of predicting the position of a body moving under an inverse-square central force—like a planet orbiting the Sun—as a function of time. Describing the motion in a plane, the two-dimensional Lagrangian in Cartesian coordinates is

$$L(x, y, \dot{x}, \dot{y}) = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}m\dot{y}^2 + \frac{k}{(x^2 + y^2)^{1/2}}. \quad (1)$$

The potential energy is negative, which represents an attractive force towards the center, which can be a massive celestial object ($k = GMm$) or an atomic nucleus ($k = Ze^2(4\pi\epsilon_0)^{-1}$). The canonical momenta are

$$p_x = \frac{\partial L}{\partial \dot{x}} = m\dot{x}, \quad p_y = \frac{\partial L}{\partial \dot{y}} = m\dot{y}, \quad (2)$$

from which we can invert to find the speeds in terms of the canonical momenta $\dot{x} = p_x/m$, $\dot{y} = p_y/m$, needed to construct the Hamiltonian

$$\begin{aligned} H(x, y, p_x, p_y) &= \sum_k p_k \dot{q}_k - L(q, \dot{q}(p)) \\ &= p_x \dot{x} + p_y \dot{y} - L(q, \dot{q}(p)) \\ &= \frac{p_x^2}{m} + \frac{p_y^2}{m} - \frac{1}{2}m \left(\frac{p_x}{m}\right)^2 - \frac{1}{2}m \left(\frac{p_y}{m}\right)^2 - \frac{k}{(x^2 + y^2)^{1/2}} \\ &= \frac{p_x^2}{2m} + \frac{p_y^2}{2m} - \frac{k}{(x^2 + y^2)^{1/2}}. \end{aligned} \quad (3)$$

Since this Hamiltonian does not explicitly depend on time, the total energy of the system is conserved

$$\frac{dH}{dt} = \frac{\partial H}{\partial t} = 0. \quad (4)$$

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The Hamilton equations

$$\dot{q}_k = \frac{\partial H}{\partial p_k}, \quad \dot{p}_k = -\frac{\partial H}{\partial q_k} \quad (5)$$

lead to the following equations of motion

$$\begin{aligned} \dot{x} &= \frac{\partial H}{\partial p_x} = \frac{p_x}{m}, & \dot{p}_x &= -\frac{\partial H}{\partial x} = -\frac{kx}{(x^2 + y^2)^{3/2}}, \\ \dot{y} &= \frac{\partial H}{\partial p_y} = \frac{p_y}{m}, & \dot{p}_y &= -\frac{\partial H}{\partial y} = -\frac{ky}{(x^2 + y^2)^{3/2}}, \end{aligned} \quad (6)$$

which exhibit a complicated form, with the time evolution of each component of the canonical momentum depending on both x and y . From here it is not obvious to identify if there is any other constant of motion (conserved quantity) that could help us simplifying the equations of motion. Let us now introduce the transformation that takes Cartesian coordinates into polar coordinates:

$$\begin{aligned} x &\rightarrow x(r, \theta) = r \cos \theta, \\ y &\rightarrow y(r, \theta) = r \sin \theta. \end{aligned} \quad (7)$$

The corresponding velocity in each axis in terms of the new coordinates become

$$\begin{aligned} \dot{x} &= \frac{dx}{dt} = \dot{r} \cos \theta - r \dot{\theta} \sin \theta, \\ \dot{y} &= \frac{dy}{dt} = \dot{r} \sin \theta + r \dot{\theta} \cos \theta, \end{aligned} \quad (8)$$

which we can directly replace into the Lagrangian

$$\begin{aligned} L'(r, \theta, \dot{r}, \dot{\theta}) &= \frac{1}{2}m(\dot{r} \cos \theta - r \dot{\theta} \sin \theta)^2 + \frac{1}{2}m(\dot{r} \sin \theta + r \dot{\theta} \cos \theta)^2 + \frac{k}{r} \\ &= \frac{1}{2}m(\dot{r}^2 \cos^2 \theta - 2r\dot{r}\dot{\theta} \cos \theta \sin \theta + r^2\dot{\theta}^2 \sin^2 \theta \\ &\quad + \dot{r}^2 \sin^2 \theta + 2r\dot{r}\dot{\theta} \cos \theta \sin \theta + r^2\dot{\theta}^2 \cos^2 \theta) + \frac{k}{r} \\ &= \left(\frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\dot{\theta}^2 \right) \underbrace{(\cos^2 \theta + \sin^2 \theta)}_1 + \frac{k}{r} \\ &= \frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\dot{\theta}^2 + \frac{k}{r}. \end{aligned} \quad (9)$$

The canonical momenta now are

$$p_r = \frac{\partial L}{\partial \dot{r}} = m\dot{r}, \quad p_\theta = \frac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta} \quad (10)$$

from which we can invert to find the speeds in terms of the canonical momenta $\dot{r} = p_r/m$, $\dot{\theta} = p_\theta/(mr^2)$, needed to construct the Hamiltonian

$$\begin{aligned}
H'(r, \theta, p_r, p_\theta) &= \sum_k p_k \dot{q}_k - L(q, \dot{q}(p)) \\
&= p_r \dot{r} + p_\theta \dot{\theta} - L(q, \dot{q}(p)) \\
&= \frac{p_r^2}{m} + \frac{p_\theta^2}{mr^2} - \frac{p_r^2}{2m} - \frac{p_\theta^2}{2mr^2} - \frac{k}{r} \\
&= \frac{p_r^2}{2m} + \frac{p_\theta^2}{2mr^2} - \frac{k}{r}.
\end{aligned} \tag{11}$$

Here we notice that the canonical variable θ does not explicitly appear in the Hamiltonian. We call θ a cyclic variable, which implies that its canonical momentum p_θ is a constant of motion. This is the celebrated conservation of angular momentum, which happens for any central potential (potential that only depends on the distance to the origin). The Hamilton equations lead to the following equations of motion

$$\begin{aligned}
\dot{r} &= \frac{\partial H'}{\partial p_r} = \frac{p_r}{m}, & \dot{p}_r &= -\frac{\partial H'}{\partial r} = \frac{p_\theta^2}{mr^3} - \frac{k}{r^2}, \\
\dot{\theta} &= \frac{\partial H'}{\partial p_\theta} = \frac{p_\theta}{mr^2}, & \dot{p}_\theta &= -\frac{\partial H'}{\partial \theta} = 0.
\end{aligned} \tag{12}$$

Here we can confirm that the angular momentum p_θ is a constant of motion. The equation for p_r involves a single variable and the problem is significantly simplified.

2 Jacobi's strategy for solving Hamiltonian systems

In 1837, just a few years after Hamilton published his formulation of classical mechanics [2], Carl Gustav Jacobi published [3] a paper laying the basis foundations for a clever method for solving Hamiltonian systems. The idea is quite straightforward. Beginning with a system characterized by canonical variables q, p and a Hamiltonian $H(q, p)$. Jacobi proposed to find a transformation of q and p to a new set of positions and momenta Q and P and the corresponding new Hamiltonian $H'(Q, P)$ so that the form of the Hamilton equations is preserved

$$\begin{aligned}
\dot{q} &= \frac{\partial H}{\partial p} & \rightarrow & & \dot{Q} &= \frac{\partial H'}{\partial P} \\
\dot{p} &= -\frac{\partial H}{\partial q} & \rightarrow & & \dot{P} &= -\frac{\partial H'}{\partial Q}
\end{aligned} \tag{13}$$

The purpose of this transformation is to find a new way of describing the physical system in coordinates that allow a simple solution of the new equations of

motion. After the system is solved, the inverse transformation can be used to obtain the original coordinates back.

Strategy for solving a Hamiltonian system

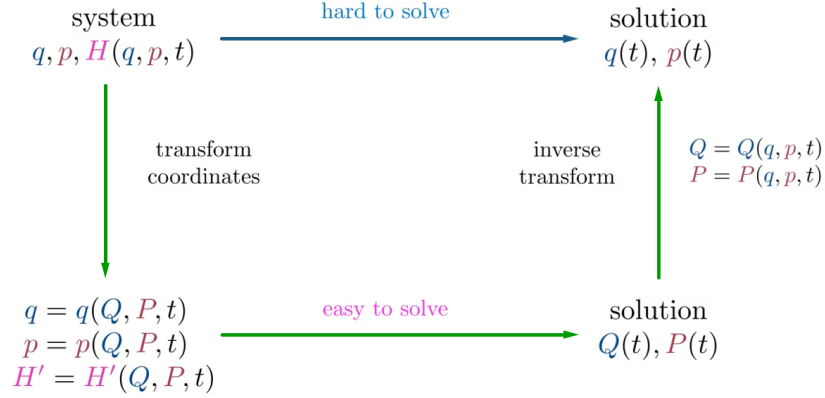


Figure 1: Instead of solving a complicated set of equations, Jacobi's method takes a detour by transforming the system into a simple problem, solving it, and then transforming back to the original variables.

2.1 Simplifying the simple harmonic oscillator

In one dimension, the Lagrangian of the simple harmonic oscillator is

$$L(q, \dot{q}) = \frac{1}{2}m\dot{q}^2 - \frac{1}{2}m\omega^2 q^2. \quad (14)$$

The Euler-Lagrange equation becomes

$$\begin{aligned} \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} &= 0 \\ \frac{d}{dt}(m\dot{q}) - (-m\omega^2 q) &= 0 \\ \ddot{q} + \omega^2 q &= 0, \end{aligned} \quad (15)$$

whose solution is given by

$$q(t) = A \sin(\omega t + \phi), \quad (16)$$

where the amplitude A and the phase ϕ are integration constants. Using the Hamiltonian formalism, instead, the Hamiltonian is

$$H(q, p) = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2, \quad (17)$$

and the Hamilton equations become

$$\dot{q} = \frac{\partial H}{\partial p} = \frac{p}{m}, \quad \dot{p} = -\frac{\partial H}{\partial q} = -m\omega^2 q. \quad (18)$$

Taking the time derivative of the first equation and replacing $\dot{p}$ from the second, we get

$$\ddot{q} = \frac{\dot{p}}{m} = -\omega^2 q, \quad (19)$$

which is the same as Eq. (15). Since the Hamiltonian (17) characterizes the total energy of the oscillator, we can write

$$H(q, p) = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2 = E. \quad (20)$$

At the maximum displacement of the oscillator, which defines its amplitude $q = A$, the oscillator is at rest ($p = 0$); hence, evaluating the Hamiltonian (20) at this point on the oscillator's motion we get

$$H(A, 0) = \frac{1}{2}m\omega^2 A^2 = E, \quad (21)$$

from where we can write the oscillation amplitude in terms of the total energy of the oscillator

$$A = \sqrt{\frac{2E}{m\omega^2}}, \quad (22)$$

and the general solution (16) becomes

$$q(t) = \sqrt{\frac{2E}{m\omega^2}} \sin(\omega t + \phi). \quad (23)$$

Let us now consider the following transformation of the canonical variables

$$\begin{aligned} q &\rightarrow q(Q, P) = \sqrt{\frac{2P}{m\omega}} \sin Q, \\ p &\rightarrow p(Q, P) = \sqrt{2m\omega P} \cos Q. \end{aligned} \quad (24)$$

Replacing these relations into the Hamiltonian (17) we get the following

$$\begin{aligned} H(q, p) &\rightarrow H'(Q, P) = \frac{1}{2m} (\sqrt{2m\omega P} \cos Q)^2 + \frac{1}{2}m\omega^2 \left(\sqrt{\frac{2P}{m\omega}} \sin Q \right)^2 \\ &= \omega P (\underbrace{\cos^2 Q + \sin^2 Q}_1) \\ &= \omega P \end{aligned} \quad (25)$$

From this extremely simple Hamiltonian, the Hamilton equations become

$$\dot{Q} = \frac{\partial H'}{\partial P} = \omega, \quad \dot{P} = 0. \quad (26)$$

The equation for P can be directly solved setting the Hamiltonian (25) equal to the total energy E of the oscillator, which gives the constant value

$$P = \frac{E}{\omega}. \quad (27)$$

The differential equation for Q can be directly solved as

$$Q(t) = \omega t + Q_0, \quad (28)$$

where Q_0 is an integration constant. Following Jacobi's strategy (Fig. 1), we replace the solutions for Q, P into the transformation (24), which directly gives the solution (23) for the simple harmonic oscillator

$$\begin{aligned} q &= \sqrt{\frac{2E}{m\omega^2}} \sin(\omega t + Q_0), \\ p &= \sqrt{2m\omega E} \cos(\omega t + Q_0). \end{aligned} \quad (29)$$

This example shows the application of Jacobi's strategy (Fig. 1) to easily solve a Hamiltonian system.

3 Canonical Transformations

In the development of the method showcased above, Jacobi found that making the new canonical variables Q, P obey equations of motion preserving the form of the original q, p Hamilton equations for the new Hamiltonian H' , as shown in Eq. (13), also preserve the geometric structure of phase space [4]. Transformations that satisfy this condition are called *canonical transformations*

$$\begin{aligned} q &= q(Q, P), & Q &= Q(q, p), \\ p &= p(Q, P), & P &= P(q, p). \end{aligned} \quad (30)$$

In particular, a very important structure that is preserved is the Poisson bracket [5]. Discussed in detail in video 41 [5], the Poisson bracket serves as way to generate the time evolution of any function in phase space. Under a canonical transformation not only the structure but also the value of Poisson brackets remains unchanged.

In the original coordinates q, p , the canonical variables satisfy

$$\begin{aligned} \{q_j, p_k\}_{q,p} &= \delta_{jk}, \\ \{q_j, q_k\}_{q,p} &= 0, \\ \{p_j, p_k\}_{q,p} &= 0. \end{aligned} \quad (31)$$

An equivalent set of canonical Poisson brackets can be written for the canonical variables Q, P after the canonical transformation

$$\begin{aligned} \{Q_j, P_k\}_{Q,P} &= \delta_{jk}, \\ \{Q_j, Q_k\}_{Q,P} &= 0, \\ \{P_j, P_k\}_{Q,P} &= 0. \end{aligned} \quad (32)$$

Consider two general functions u and v in phase space. In the original coordinates q, p , the Poisson bracket is given by

$$\{u, v\}_{q,p} = \sum_k \left(\frac{\partial u}{\partial q_k} \frac{\partial v}{\partial p_k} - \frac{\partial v}{\partial q_k} \frac{\partial u}{\partial p_k} \right), \quad (33)$$

where the sum is over the N degrees of freedom $k \in \{1, \dots, N\}$. Similarly, in the new coordinates, the Poisson bracket is

$$\{u, v\}_{Q,P} = \sum_k \left(\frac{\partial u}{\partial Q_k} \frac{\partial v}{\partial P_k} - \frac{\partial v}{\partial Q_k} \frac{\partial u}{\partial P_k} \right). \quad (34)$$

The invariance of the Poisson bracket under canonical transformations implies

$$\{u, v\}_{q,p} = \{u, v\}_{Q,P}. \quad (35)$$

Exercise: demonstrate that the Poisson bracket is invariant under canonical transformations (a step-by-step solution will be posted in the near future).

The equations of motion satisfied by the two sets of coordinates, implies that they can be derived from a similar action principle

$$\delta \int_{t_1}^{t_2} \left(\sum_k p_k \dot{q}_k - H \right) dt = 0, \quad \delta \int_{t_1}^{t_2} \left(\sum_k P_k \dot{Q}_k - H' \right) dt = 0. \quad (36)$$

Since these two relations are valid, it is tempting to set their integrands to be equal. However, this would be too restrictive. A more general condition would be to set the two integrands to be proportional. However, the proportionality constant could be easily eliminated by rescaling the coordinates. The most general way to relate the two integrands in Eq. 36 is by setting their difference to a total time derivative, which does not affect the action and keeps it stationary. From this assessment, we can write

$$\sum_k p_k \dot{q}_k - H = \sum_k P_k \dot{Q}_k - H' + \frac{dF}{dt}. \quad (37)$$

The function F is called the *generating function* of the transformation, as it will be used to generate the canonical transformation. In general, the generating function could depend on all the canonical variables q, p, Q, P (per degree of freedom) and time t . However, the transformation (30) corresponds to two constraints (per degree of freedom) and only pairs of canonical variables and time are independent. There are four so-called basic canonical transformations generated by selecting different pairs of canonical variables. See Table 1.

The fundamental ideas behind canonical transformations are as follows:

- the equations of motion of a mechanical system can be highly simplified by transforming the canonical variables q, p to new coordinates Q, P in phase space
- transformations that preserve the form of the Hamilton equations (canonical transformations) are defined by Eq. (37)
- we can design the generating function F so that the generated transformation leads to the new Hamiltonian H' of a given a desired form.

3.1 Type-1 generating function

Let us consider a type-1 generating function $F = F_1(q, Q, t)$. Since this function only depends on q, Q and t

$$\frac{dF}{dt} = \sum_k \frac{\partial F_1}{\partial q_k} \dot{q}_k + \sum_k \frac{\partial F_1}{\partial Q_k} \dot{Q}_k + \frac{\partial F_1}{\partial t}. \quad (38)$$

Replacing this expression into Eq. (37) and grouping common factors we get

$$\sum_k \left(\frac{\partial F_1}{\partial q_k} - p_k \right) \dot{q}_k + \sum_k \left(P_k + \frac{\partial F_1}{\partial Q_k} \right) \dot{Q}_k + \left(H - H' + \frac{\partial F_1}{\partial t} \right) = 0. \quad (39)$$

This equation can only be satisfied if each of the terms in parenthesis individually vanishes, which gives the so-called generating-function derivatives

$$p_k = \frac{\partial F_1}{\partial q_k}, \quad P_k = -\frac{\partial F_1}{\partial Q_k}, \quad H + \frac{\partial F_1}{\partial t} = H'. \quad (40)$$

3.1.1 Simple harmonic oscillator (again)

Consider the type-1 generating function

$$F_1(q, Q) = \frac{1}{2} m \omega q^2 \cot Q, \quad (41)$$

which does not explicitly depend on time. Using equations (40), the momentum P is¹

$$P = -\frac{\partial F_1}{\partial Q} = \frac{m \omega q^2}{2 \sin^2 Q}. \quad (42)$$

Solving this equation for q we find

$$q = \sqrt{\frac{2P}{m\omega}} \sin Q. \quad (43)$$

¹ $\frac{d}{dx} \cot x = -\csc^2 x$

Similarly, from equations (40), the momentum p is

$$p = \frac{\partial F_1}{\partial q} = m\omega q \frac{\cos Q}{\sin Q}, \quad (44)$$

where q as a function of Q and P can be replaced from the expression found above. The final result is that using the generating function given by Eq. (41) we get

$$\begin{aligned} q &\rightarrow q(Q, P) = \sqrt{\frac{2P}{m\omega}} \sin Q, \\ p &\rightarrow p(Q, P) = \sqrt{2m\omega P} \cos Q, \end{aligned} \quad (45)$$

which is exactly the transformation (24) used in Sec. 2.1 to simplify the equations of the simple harmonic oscillator.

3.2 Type-2 generating function

Let us consider a type-2 generating function $F = F_2(q, P, t) - \sum_k Q_k P_k$. Since this function only depends on q, P and t

$$\frac{dF}{dt} = \sum_k \frac{\partial F_2}{\partial q_k} \dot{q}_k + \sum_k \frac{\partial F_2}{\partial P_k} \dot{P}_k + \frac{\partial F_2}{\partial t} + \sum_k \dot{Q}_k P_k + \sum_k Q_k \dot{P}_k. \quad (46)$$

Replacing this expression into Eq. (37) and grouping common factors we get

$$\sum_k \left(\frac{\partial F_2}{\partial q_k} - p_k \right) \dot{q}_k + \sum_k \left(\frac{\partial F_2}{\partial P_k} - Q_k \right) \dot{P}_k + \left(H - H' + \frac{\partial F_2}{\partial t} \right) = 0. \quad (47)$$

This equation can only be satisfied if each of the terms in parenthesis individually vanishes, which gives the so-called generating-function derivatives

$$p_k = \frac{\partial F_2}{\partial q_k}, \quad Q_k = \frac{\partial F_2}{\partial P_k}, \quad H + \frac{\partial F_1}{\partial t} = H'. \quad (48)$$

The same procedure can be followed for deriving the equations for generating functions of type 3 and 4, summarized in Table 1.

3.3 The bridge to quantum theory

Canonical transformations are deeply connected to the early development of quantum theory. A particular way of constructing a type-1 generating function lays at the roots of the Bohr–Sommerfeld quantization rule (see video 21 [6])

$$J = \oint p dq = nh, \quad (49)$$

that guided old quantum theory since the development of Bohr’s atomic model (see video 17 [7]), and the Born–Kramers transformation

$$\alpha \frac{d\Phi(n, \alpha)}{dJ} \longrightarrow \frac{\Phi_{n+\alpha, n} - \Phi_{n, n-\alpha}}{h}, \quad (50)$$

Generating Function	Generating Function Derivatives
$F = F_1(q, Q, t)$	$p_k = \frac{\partial F_1}{\partial q_k} \quad P_k = -\frac{\partial F_1}{\partial Q_k}$
$F = F_2(q, P, t) - \sum_k Q_k P_k$	$p_k = \frac{\partial F_2}{\partial q_k} \quad Q_k = \frac{\partial F_2}{\partial P_k}$
$F = F_3(p, Q, t) + \sum_k q_k p_k$	$q_k = -\frac{\partial F_3}{\partial p_k} \quad P_k = -\frac{\partial F_3}{\partial Q_k}$
$F = F_4(p, P, t) + \sum_k q_k p_k - \sum_k Q_k P_k$	$q_k = -\frac{\partial F_4}{\partial p_k} \quad Q_k = \frac{\partial F_4}{\partial P_k}$

Table 1: The four basic canonical transformations.

that guided Heisenberg and the development of matrix mechanics (see videos [8] and [9]). Intended to generate a new Hamiltonian in which one of the canonical variables is cyclic

$$H(q, p) \rightarrow H'(Q, P) = H(P), \quad (51)$$

this type of canonical transformation makes the new canonical momentum P a constant of motion, which ideal for mechanics systems that exhibit a multi-periodic motion. This is the idea behind **action-angle variables**.

Similarly, an extreme simplification of the equations of motion can be achieved by constructing a type-2 generating function that makes the new Hamiltonian identically zero

$$H(q, p) \rightarrow H'(Q, P) = 0. \quad (52)$$

this type of canonical transformation makes the all new canonical variables constants of motion. This is the idea behind **Hamilton–Jacobi theory**. In this case, the unknown generating function $F_2(q, P, t)$ is referred to as *Hamilton’s Principal Function* and denoted by S (there is a profound reason for choosing this notation). In the particular case of a time-independent Hamiltonian H , it can be shown that the generating function is given by

$$S(q, t) = W(q) - Et, \quad (53)$$

where the variable P is implicit as an integration constant and the function $W(q)$ is called *Hamilton’s Characteristic Function*. Using Eq. (48) we find

$$p = \frac{\partial S}{\partial q}, \quad H + \frac{\partial S}{\partial t} = 0. \quad (54)$$

Since Eq. (53) implies that $\partial S / \partial t = -E$, the last result implies

$$H\left(q, \frac{\partial S}{\partial q}\right) = E. \quad (55)$$

This is called the time-independent Hamilton–Jacobi equation. This is the starting point in Schrödinger’s first paper on wave mechanics.

Comments

These notes will be updated in case that any correction is needed. If you spot any typo or error in the calculation just use the comments section of the video and report the problem, I will happily fix it here.

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